

GSERM - St. Gallen 2026

Analyzing Panel Data

June 23, 2026

Two-Way Variation

Two-way variation:

$$Y_{it} = \mathbf{X}_{it}\beta + \gamma V_i + \delta W_t + u_{it}$$

where V_i are the ℓ predictors which don't vary over time (within a unit), and W_t are the m predictors which don't vary across units (for a given time point).

Note that we can write:

$$\alpha_i = \sum_{\ell} (\gamma V_i)$$

and

$$\eta_t = \sum_m (\delta W_t).$$

So:

$$\begin{aligned} Y_{it} &= \mathbf{X}_{it}\beta + \gamma V_i + \delta W_t + u_{it} \\ &= \mathbf{X}_{it}\beta + \alpha_i + \eta_t + u_{it} \end{aligned}$$

“One-Way” and “Two-Way” Effects

“Two-way” effects:

$$Y_{it} = \mathbf{X}_{it}\boldsymbol{\beta} + \alpha_i + \eta_t + u_{it}$$

“One-way” effects:

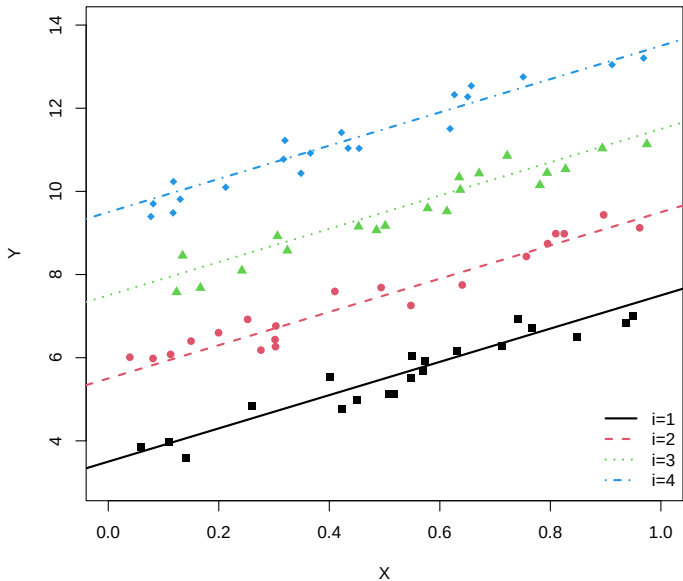
- Assuming $\alpha_i = 0$:

$$Y_{it} = \mathbf{X}_{it}\boldsymbol{\beta} + \eta_t + u_{it} \quad (\text{“time effects”})$$

- Assuming $\eta_t = 0$:

$$Y_{it} = \mathbf{X}_{it}\boldsymbol{\beta} + \alpha_i + u_{it} \quad (\text{“unit effects”})$$

Intuition: One-Way Unit Effects



(One-Way) “Fixed” Effects

“Brute force” model fits:

$$\begin{aligned} Y_{it} &= \mathbf{X}_{it}\beta_{FE} + \alpha_i + u_{it} \\ &= \mathbf{X}_{it}\beta_{FE} + \alpha_1 I(i = 1)_i + \alpha_2 I(i = 2)_i + \dots + u_{it} \end{aligned}$$

In other words:

- Specify $\mathbf{X}_{it}$,
- Fit a model that includes both $\mathbf{X}_{it}\beta$ and N (or $N - 1$) indicator variables (with parameters α_i), one for each unit i
 - If the model includes an intercept (β_0), then one unit is omitted and becomes the “reference” category
 - If the model omits an intercept, then all N $\hat{\alpha}$ s can be estimated; $\hat{\alpha}_i$ is then the expected value of Y for unit i when all values of $\mathbf{X}_{it}$ are zero

“Fixed” Effects (continued)

Alternatively, decompose:

$$\bar{\mathbf{X}}_i = \frac{\sum_{N_i} \mathbf{X}_{it}}{N_i}$$

and

$$\tilde{\mathbf{X}}_{it} = \mathbf{X}_{it} - \bar{\mathbf{X}}_i.$$

Yields:

$$Y_{it} = \bar{\mathbf{X}}_i \beta_B + \tilde{\mathbf{X}}_{it} \beta_W + \alpha_i + u_{it}$$

But!

$$\text{corr}(\bar{\mathbf{X}}_i \beta_B, \alpha_i) = 1.0$$

“Fixed” Effects = “Within” Effects

This means that:

$$\begin{aligned}Y_{it}^* &= Y_{it} - \bar{Y}_i \\ \mathbf{X}_{it}^* &= \mathbf{X}_{it} - \bar{\mathbf{X}}_i\end{aligned}$$

gives:

$$Y_{it}^* = \mathbf{X}_{it}^* \beta_{FE} + u_{it}.$$

→ **A “fixed effects” model is actually a “within-effects” model.**

- “Fixed effects” models only use the *within-unit* variation in Y and $\mathbf{X}$
- $\hat{\beta}$ s for variables that do not vary within units cannot be estimated

“Fixed” Effects: Test(s)

The one-way FE model implies that $\alpha_i \neq \alpha_j$ for at least some $i \neq j$. We can test this assumption using a standard F -test of the hypothesis:

$$H_0 : \alpha_i = \alpha_j \forall i \neq j$$

versus

$$H_A : \alpha_i \neq \alpha_j \text{ for some } i \neq j$$

Specifically, the test statistic

$$F = \frac{(\text{SSE}_{OLS} - \text{SSE}_{FE}) / N - 1}{\text{SSE}_{FE} / [NT - (N - 1)]}$$

is $\sim F_{N-1, NT-(N-1)}$.

Example Redux: WDI, 1960-2025

The World Development Indicators

- Cross-national country-level time series data
- $N = 215$ countries, $T = 65$ years (1960-2025) + missingness
- Variables:
 - Geography: land area, arable land
 - Population indicators
 - Demographics: Birth rates, life expectancy, etc.
 - Economics: GDP, inflation, trade, FDI, etc.
 - Governments: expenditures, policies, etc.
- Full descriptions are listed in the Github repo [here](#).

Data Summary

```
> describe(wdi,fast=TRUE,ranges=FALSE,check=TRUE)
```

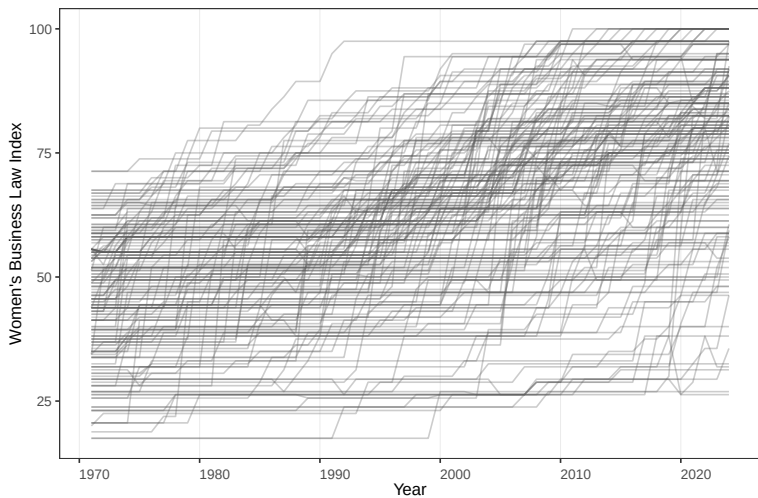
	vars	n	mean	sd	skew	kurtosis	se
ISO3	1	14298	NaN	NA	NA	NA	NA
Year	2	14298	1992.54	1.9e+01	0.00	-1.20	0.16
Region	3	14124	NaN	NA	NA	NA	NA
country	4	14190	NaN	NA	NA	NA	NA
iso3c	5	14190	NaN	NA	NA	NA	NA
LandArea	6	12288	596587.32	1.6e+06	5.46	34.81	14839.81
ArablePercent	7	11871	13.40	1.4e+01	1.47	1.96	0.13
Population	8	13945	25537942.47	1.1e+08	9.72	105.61	898698.32
PopGrowth	9	13728	1.74	1.8e+00	0.73	21.23	0.02
RuralPopulation	10	13975	47.79	2.6e+01	-0.09	-0.99	0.22
UrbanPopulation	11	13975	52.21	2.6e+01	0.09	-0.99	0.22
BirthRatePer1K	12	13945	27.51	1.3e+01	0.25	-1.23	0.11
FertilityRate	13	13943	3.80	2.0e+00	0.45	-1.17	0.02
PrimarySchoolAge	14	11539	6.13	6.1e-01	-0.03	0.13	0.01
LifeExpectancy	15	13941	65.31	1.1e+01	-0.77	0.19	0.10
AgeDepRatioOld	16	13945	10.84	7.3e+00	1.80	5.05	0.06
CO2Emissions	17	11165	5.04	1.1e+01	8.69	102.26	0.11
GDP	18	11307	243922334825.73	1.2e+12	11.54	161.13	10845501840.29
GDPPerCapita	19	11307	12443.80	1.9e+04	3.17	15.21	182.91
GDPPerCapGrowth	20	11184	1.91	6.5e+00	1.68	40.87	0.06
Inflation	21	8949	22.85	3.2e+02	54.29	3652.65	3.38
TotalTrade	22	8802	78.15	5.4e+01	3.01	17.96	0.57
Exports	23	8802	36.40	2.9e+01	2.98	16.46	0.31
Imports	24	8811	41.76	2.7e+01	2.52	13.58	0.29
FDIIn	25	9198	5.46	4.4e+01	15.43	561.70	0.46
AgriEmployment	26	6496	27.25	2.3e+01	0.68	-0.64	0.29
NetAidReceived	27	9218	528849286.43	1.1e+09	11.21	269.53	11599881.51
MobileCellSubscriptions	28	10587	39.15	5.4e+01	1.26	1.61	0.52
NaturalResourceRents	29	9211	6.84	1.1e+01	2.60	8.04	0.12
MilitaryExpenditures	30	7892	2.68	3.2e+00	9.43	238.04	0.04
GovtExpenditures	31	8470	16.39	8.3e+00	3.65	32.91	0.09
PublicEdExpend	32	5125	4.29	2.1e+00	6.10	158.28	0.03
PublicHealthExpend	33	4547	3.31	2.4e+00	1.37	3.43	0.04
HIVDeaths	34	5020	7002.28	2.1e+04	5.74	45.55	294.03
PostColdWar	35	14190	0.55	5.0e-01	-0.18	-1.97	0.00
WomenBusLawIndex	36	10260	59.97	1.9e+01	0.01	-0.59	0.19
PaidParentalLeave	37	10260	0.11	3.1e-01	2.50	4.24	0.00

WDI's Women, Business and the Law Index (WBLI)

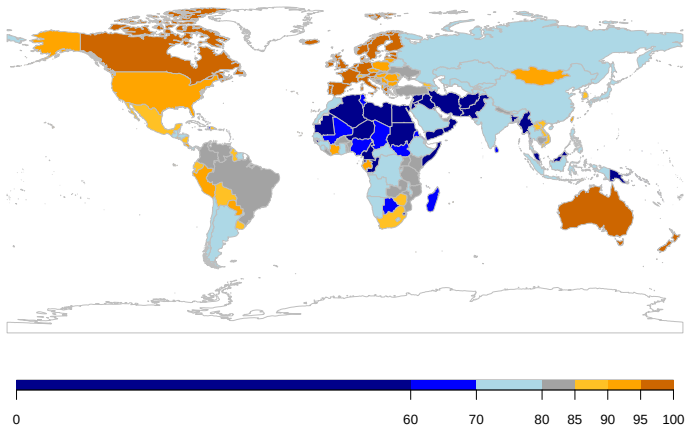
The basis for World Bank reports in 2021 (and 2026)...

- Version 1.0 examines “the laws and regulations that affect women's economic opportunity in 190 economies” from 1971-2024.
- An index comprising eight indicators “structured around women's interactions with the law as they move through their careers: *Mobility, Workplace, Pay, Marriage, Parenthood, Entrepreneurship, Assets, and Pension.*”
- The WBL Index:
 - Theoretically ranges from 0 - 100
 - In practice: lowest values ≈ 18
 - Higher values correspond to higher levels of women's empowerment and greater opportunities and support for women, particularly in business
- “Better performance in the areas measured by the *Women, Business and the Law Index* is associated with a more narrow gender gap in development outcomes, higher female labor force participation, lower vulnerable employment, and greater representation of women in national parliaments.”

Trend Line Visualization (using panelView)



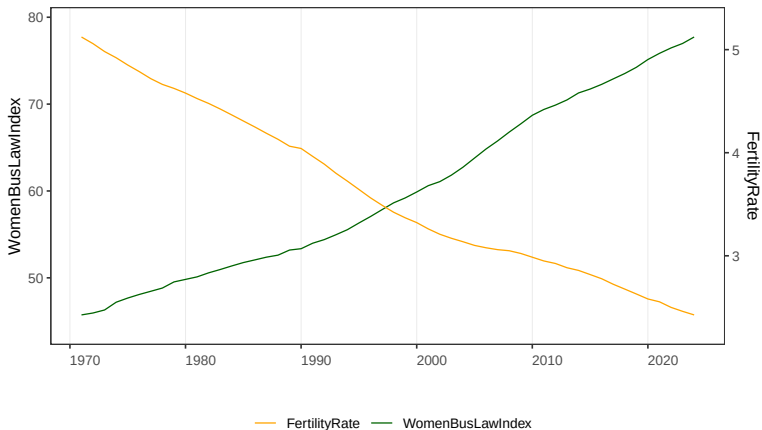
Map! WBLI (2023)



Categorical Variable Visualization (using panelView)



Bivariate Visualization (using panelView)



`plm` (panel linear models) is the workhorse R package for fitting linear regression models to panel data. To do so, you must first declare the data as panel data and indicate the variables containing the unit (i) and period (t) indices:

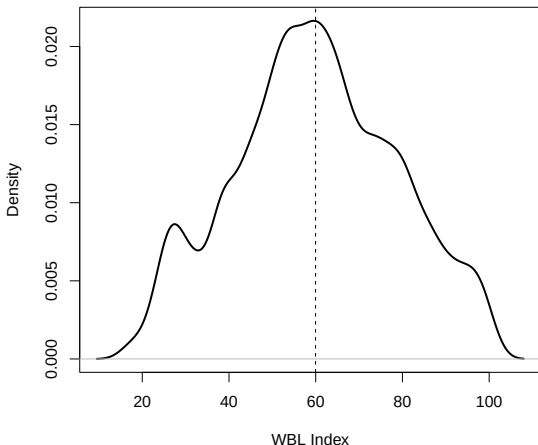
```
> WDI<-pdata.frame(wdi,index=c("ISO3","Year"))
> class(WDI)
[1] "pdata.frame" "data.frame"

> WBLI<-WDI$WomenBusLawIndex
> class(WBLI)
[1] "pseries" "numeric"
```


WBLI: Total Variation

```
> describe(WBLI,na.rm=TRUE) # all variation
```

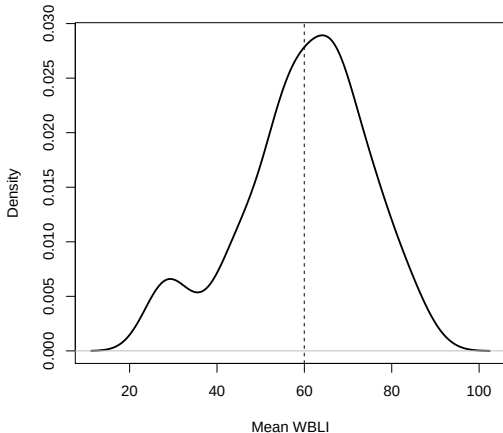
	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
X1	1	10260	60	19	59	60	19	18	100	82	0.01	-0.59	0.19



WBLI: "Between" Variation

```
> describe(plm::between(WBLI,effect="individual",na.rm=TRUE)) # "between" variation
```

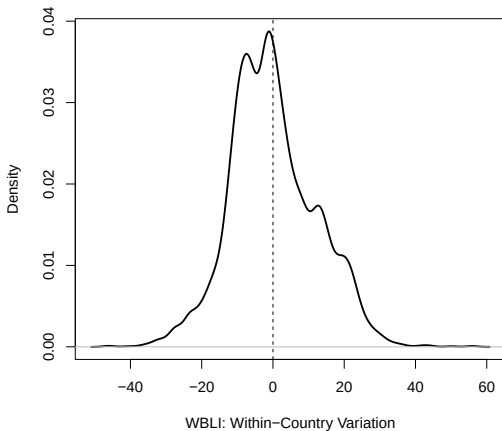
	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
X1	1	190	60	14	61	61	13	24	90	66	-0.51	-0.11	1



WBLI: “Within” Variation

```
> describe(Within(WBLI,na.rm=TRUE)) # "within" variation
```

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
X1	1	10260	0	12	-1.2	-0.33	11	-46	56	102	0.26	0.32	0.12



A Regression Model

Regression model:

$$\text{WBLI}_{it} = \beta_0 + \beta_1 \text{Population Growth}_{it} + \beta_2 \text{Urban Population}_{it} + \beta_3 \text{Fertility Rate}_{it} + \beta_4 \ln(\text{GDP Per Capita})_{it} + \beta_5 \text{Natural Resource Rents}_{it} + \beta_6 \text{Post Cold War}_t + u_{it}$$

$$\begin{aligned} \text{Expectations: } \beta_1 &< 0 & \beta_3 &< 0 & \beta_5 &< 0 \\ \beta_2 &> 0 & \beta_4 &> 0 & \beta_6 &> 0 \end{aligned}$$

Descriptive Statistics:

	vars	n	mean	sd	median	min	max	range	skew	kurtosis	se
WomenBusLawIndex	1	8252	60.0	18.85	60.6	17.50	100.0	82.5	-0.02	-0.66	0.21
PopGrowth	2	8252	1.7	1.67	1.6	-27.47	21.7	49.2	-0.09	30.12	0.02
UrbanPopulation	3	8252	52.5	23.88	53.0	3.01	100.0	97.0	0.03	-1.05	0.26
FertilityRate	4	8252	3.6	1.89	3.1	0.77	8.6	7.8	0.53	-1.01	0.02
NaturalResourceRents	5	8252	7.3	11.18	2.5	0.00	88.6	88.6	2.49	7.40	0.12
PostColdWar	6	8252	0.7	0.46	1.0	0.00	1.0	1.0	-0.87	-1.24	0.01
lnGDPPerCap	7	8252	8.3	1.46	8.2	4.93	11.7	6.7	0.11	-0.89	0.02

Where is the Variation in our Data?

Variable	Dimension	Mean	SD	Min	Max	Observations
Women Business Law Index	overall	60.033	18.849	17.5	100	N = 8252
	between		15.236	23.173	89.361	n = 188
	within		11.177	15.182	118.692	T = 43.894
Population Growth	overall	1.666	1.673	-27.471	21.7	N = 8252
	between		1.27	-1.077	6.328	n = 188
	within		1.143	-29.138	17.381	T = 43.894
Urban Population	overall	52.478	23.876	3.01	100	N = 8252
	between		22.723	8.472	100	n = 188
	within		7.073	15.191	85.194	T = 43.894
Fertility Rate	overall	3.59	1.887	0.772	8.606	N = 8252
	between		1.662	1.292	7.588	n = 188
	within		0.916	0.874	6.98	T = 43.894
Natural Resource Rents	overall	7.284	11.184	0	88.592	N = 8252
	between		10.06	0	43.634	n = 188
	within		5.279	-23.94	61.391	T = 43.894
Post-Cold War	overall	0.7	0.458	0	1	N = 8252
	between		0.154	0.345	1	n = 188
	within		0.439	-0.27	1.355	T = 43.894
ln(GDP Per Capita)	overall	8.34	1.457	4.926	11.671	N = 8252
	between		1.394	5.682	11.199	n = 188
	within		0.357	6.528	10.257	T = 43.894

Regression: Pooled OLS

```
> OLS<-plm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+  
+          log(GDPPerCapita)+NaturalResourceRents+PostColdWar,  
+          data=WDI,model="pooling")
```

```
> summary(OLS)
```

Pooling Model

Call:

```
plm(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +  
    FertilityRate + log(GDPPerCapita) + NaturalResourceRents +  
    PostColdWar, data = WDI, model = "pooling")
```

Unbalanced Panel: n = 188, T = 8-51, N = 8252

Residuals:

Min.	1st Qu.	Median	3rd Qu.	Max.
-78.29	-8.62	1.01	9.27	45.19

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	51.9633	1.7857	29.10	< 2e-16 ***
PopGrowth	-1.7837	0.1165	-15.31	< 2e-16 ***
UrbanPopulation	-0.0749	0.0104	-7.22	5.7e-13 ***
FertilityRate	-2.9900	0.1549	-19.31	< 2e-16 ***
log(GDPPerCapita)	2.5588	0.1878	13.62	< 2e-16 ***
NaturalResourceRents	-0.3506	0.0150	-23.32	< 2e-16 ***
PostColdWar	9.8824	0.3659	27.01	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 2930000

Residual Sum of Squares: 1490000

R-Squared: 0.493

Adj. R-Squared: 0.493

F-statistic: 1337.18 on 6 and 8245 DF, p-value: <2e-16

"Fixed" (Within) Effects

```
> FE<-plm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+log(GDPPerCapita)+
+         NaturalResourceRents+PostColdWar,data=WDI,effect="individual",model="within")
>
> summary(FE)
Oneway (individual) effect Within Model
```

```
Call:
plm(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +
     FertilityRate + log(GDPPerCapita) + NaturalResourceRents +
     PostColdWar, data = WDI, effect = "individual", model = "within")
```

Unbalanced Panel: n = 188, T = 8-51, N = 8252

Residuals:

	Min.	1st Qu.	Median	3rd Qu.	Max.
	-33.766	-5.090	-0.628	4.982	55.202

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)
PopGrowth	-0.1303	0.0806	-1.62	0.11
UrbanPopulation	0.3370	0.0183	18.41	< 2e-16 ***
FertilityRate	-1.5477	0.1549	-9.99	< 2e-16 ***
log(GDPPerCapita)	8.5685	0.2895	29.60	< 2e-16 ***
NaturalResourceRents	0.0898	0.0166	5.41	0.000000066 ***
PostColdWar	6.7013	0.2860	23.43	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 1030000

Residual Sum of Squares: 495000

R-Squared: 0.52

Adj. R-Squared: 0.508

F-statistic: 1454.48 on 6 and 8058 DF, p-value: <2e-16

A Nicer Table

Table: Models of WBLI

	OLS	FE
Population Growth	-1.784*** (0.117)	-0.130 (0.081)
Urban Population	-0.075*** (0.010)	0.337*** (0.018)
Fertility Rate	-2.990*** (0.155)	-1.548*** (0.155)
ln(GDP Per Capita)	2.559*** (0.188)	8.569*** (0.289)
Natural Resource Rents	-0.351*** (0.015)	0.090*** (0.017)
Post-Cold War	9.882*** (0.366)	6.701*** (0.286)
Constant	51.960*** (1.786)	
Observations	8,252	8,252
R ²	0.493	0.520
Adjusted R ²	0.493	0.508
F Statistic	1,337*** (df = 6; 8245)	1,454*** (df = 6; 8058)

*p<0.1; **p<0.05; ***p<0.01

One-Way Period Fixed Effects

The model is:

$$Y_{it} = \mathbf{X}_{it}\boldsymbol{\beta} + \eta_t + u_{it}$$

which is estimated via “demeaning” data at each time point:

$$\begin{aligned} Y_{it}^{**} &= Y_{it} - \bar{Y}_t \\ \mathbf{X}_{it}^{**} &= \mathbf{X}_{it} - \bar{\mathbf{X}}_t \end{aligned}$$

then fitting the OLS model:

$$Y_{it}^{**} = \beta_{FE} \mathbf{X}_{it}^{**} + u_{it}.$$

Comparison: Unit vs. Time Fixed Effects

Table: FE Models of WBLI (Units vs. Periods)

	OLS	FE.Units	FE.Time
Population Growth	-1.784*** (0.117)	-0.130 (0.081)	-1.928*** (0.110)
Urban Population	-0.075*** (0.010)	0.337*** (0.018)	-0.078*** (0.010)
Fertility Rate	-2.990*** (0.155)	-1.548*** (0.155)	-2.117*** (0.150)
ln(GDP Per Capita)	2.559*** (0.188)	8.569*** (0.289)	2.946*** (0.178)
Natural Resource Rents	-0.351*** (0.015)	0.090*** (0.017)	-0.382*** (0.014)
Post-Cold War	9.882*** (0.366)	6.701*** (0.286)	
Constant	51.960*** (1.786)		
Observations	8,252	8,252	8,252
R ²	0.493	0.520	0.399
Adjusted R ²	0.493	0.508	0.395
F Statistic	1,337*** (df = 6; 8245)	1,454*** (df = 6; 8058)	1,089*** (df = 5; 8196)

*p<0.1; **p<0.05; ***p<0.01

As we noted previously, the specification:

$$Y_{it} = \mathbf{X}_{it}\beta + \alpha_i + u_{it}$$

...suggests that we can use an F -test to examine the hypothesis:

$$H_0 : \alpha_i = \alpha \forall i$$

(and a similar test for $\eta_t = 0$ in the time-centered case).

Arguably better are Lagrange multiplier-based tests:

- Breusch-Pagan (1980)
- King and Wu (1997)
- See (e.g.) Croissant and Millo (2018, §4.1) for details

FE (by Unit) Model Tests

```
> pFtest(FE,OLS)
```

F test for individual effects

```
data:  WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...
F = 86, df1 = 187, df2 = 8058, p-value <2e-16
alternative hypothesis: significant effects
```

```
> plmtest(FE,effect=c("individual"),type=c("bp"))
```

Lagrange Multiplier Test - (Breusch-Pagan)

```
data:  WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...
chisq = 55538, df = 1, p-value <2e-16
alternative hypothesis: significant effects
```

```
> plmtest(FE,effect=c("individual"),type=c("kw"))
```

Lagrange Multiplier Test - (King and Wu)

```
data:  WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...
normal = 236, p-value <2e-16
alternative hypothesis: significant effects
```

FE (by Period) Model Tests

```
> pFtest(FE.Time,OLS)
```

F test for time effects

```
data:  WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...  
F = 22, df1 = 49, df2 = 8196, p-value <2e-16  
alternative hypothesis: significant effects
```

```
> plmtest(FE.Time,effect=c("time"),type=c("bp"))
```

Lagrange Multiplier Test - time effects (Breusch-Pagan)

```
data:  WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...  
chisq = 8490, df = 1, p-value <2e-16  
alternative hypothesis: significant effects
```

```
> plmtest(FE.Time,effect=c("time"),type=c("kw"))
```

Lagrange Multiplier Test - time effects (King and Wu)

```
data:  WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...  
normal = 92, p-value <2e-16  
alternative hypothesis: significant effects
```

Fixed Effects: Interpretation

FE models are *subject-specific* models, that rely entirely on *within-unit* variability to estimate variable effects.

- This means that:

$$\hat{\beta}_k = \frac{\partial E(Y|\hat{\alpha})}{\partial X_k}$$

- That is, $\hat{\beta}_k$ is the expected change in $E(Y)$ associated with a one-unit increase in observation i 's value of X_k
- Key: *within-unit* changes in $\mathbf{X}$ are associated with *within-unit* expected changes in Y .
- In a linear model, the value of $\hat{\alpha}$ doesn't affect the value of that partial derivative...

Fixed Effects: Interpretation

Mummolo and Peterson (2018) note that:

“...because the within-unit variation is always smaller (or at least, no larger) than the overall variation in the independent variable, researchers should use within-unit variation to motivate counterfactuals when discussing the substantive impact of a treatment” (2018, 829).

Significance:

- Predictors **X** in FE models typically have both cross-sectional and temporal variation
- FE models only consider *within-unit* variation in **X** and *Y*
- As a result, the degree of actual/observed (within-unit) variation in predictors is almost always less – and sometimes significantly less – than if cross-sectional variation were also considered

Interpretation Example: Urban Population

Q: How much is a “one standard deviation change in Urban Population”?

Recall its variation:

Variable	Dimension	Mean	SD	Min	Max	Observations
Urban Population	overall	52.478	23.876	3.01	100	N = 8252
	between		22.723	8.472	100	n = 188
	within		7.073	15.191	85.194	T = 43.894

“While the overall variation in the independent variable may be large, the within-unit variation used to estimate β may be much smaller” (M & P 2018, p. 830).

Whither “Fixed” Effects”?



Khoa Vu
@KhoaVuUmn



Real footage of an applied economist adding fixed effects in his regressions



Pros and Cons of “Fixed” Effects

Pros:

- Mitigates (Some) Specification Bias
- Simple + Intuitive
- Widely Used/Understood

Cons (see e.g. [Collischon and Eberl 2020](#)):

- Can't Estimate β_B
- Slowly-Changing $\mathbf{X}$ s
- (In)Efficiency / Inconsistency (Incidental Parameters)
- Cannot Control for (e.g.) Time-Varying Heterogeneity
- Sensitivity to Measurement Error

“Between” Effects

From the equation above:

$$Y_{it} = \bar{\mathbf{X}}_i \beta_B + \tilde{\mathbf{X}}_{it} \beta_W + \alpha_i + u_{it}$$

...we can derive a “Between Effects” model:

$$\bar{Y}_i = \bar{\mathbf{X}}_i \beta_B + u_{it}$$

This model:

- is essentially cross-sectional,
- is based on N observations,
- considers *only* between-unit (average) differences
- Interpretation:

$\hat{\beta}_B$ is the expected difference in Y between two units whose values on $\bar{X}$ differ by a value of 1.0.

"Between" Effects

```
> BE<-plm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+
+          log(GDPPerCapita)+NaturalResourceRents+PostColdWar,data=WDI,
+          effect="individual",model="between")
>
> summary(BE)
Oneway (individual) effect Between Model
```

```
Call:
plm(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +
      FertilityRate + log(GDPPerCapita) + NaturalResourceRents +
      PostColdWar, data = WDI, effect = "individual", model = "between")
```

```
Unbalanced Panel: n = 188, T = 8-51, N = 8252
Observations used in estimation: 188
```

```
Residuals:
    Min. 1st Qu.  Median 3rd Qu.    Max.
-27.927  -6.909   0.562   8.136  21.381
```

```
Coefficients:
              Estimate Std. Error t-value Pr(>|t|)
(Intercept)    57.3445   12.6021    4.55 0.0000098 ***
PopGrowth      -4.9889    1.1136   -4.48 0.0000132 ***
UrbanPopulation -0.0674    0.0546   -1.24  0.2184
FertilityRate   -1.5460    1.1540   -1.34  0.1820
log(GDPPerCapita) 2.1317    1.1119    1.92  0.0568 .
NaturalResourceRents -0.2998  0.0905   -3.31  0.0011 **
PostColdWar      6.5467    5.4024    1.21  0.2272
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Total Sum of Squares:    43400
Residual Sum of Squares: 18300
R-Squared:                0.579
Adj. R-Squared: 0.565
F-statistic: 41.5021 on 6 and 181 DF, p-value: <2e-16
```

A Nicer Table (Again)

Table: Models of WBLI

	OLS	FE	BE
Population Growth	-1.784*** (0.117)	-0.130 (0.081)	-4.989*** (1.114)
Urban Population	-0.075*** (0.010)	0.337*** (0.018)	-0.067 (0.055)
Fertility Rate	-2.990*** (0.155)	-1.548*** (0.155)	-1.546 (1.154)
ln(GDP Per Capita)	2.559*** (0.188)	8.569*** (0.289)	2.132* (1.112)
Natural Resource Rents	-0.351*** (0.015)	0.090*** (0.017)	-0.300*** (0.091)
Post-Cold War	9.882*** (0.366)	6.701*** (0.286)	6.547 (5.402)
Constant	51.960*** (1.786)		57.340*** (12.600)
Observations	8,252	8,252	188
R ²	0.493	0.520	0.579
Adjusted R ²	0.493	0.508	0.565
F Statistic	1,337*** (df = 6; 8245)	1,454*** (df = 6; 8058)	41.5*** (df = 6; 181)

*p<0.1; **p<0.05; ***p<0.01

“Random” Effects

Model:

$$Y_{it} = \mathbf{X}_{it}\beta + u_{it}$$

with:

$$u_{it} = \alpha_i + \lambda_t + \eta_{it}$$

and

$$\begin{aligned} E(\alpha_i) &= E(\lambda_t) = E(\eta_{it}) &= 0, \\ E(\alpha_i \lambda_t) &= E(\alpha_i \eta_{it}) = E(\lambda_t \eta_{it}) &= 0, \\ E(\alpha_i \alpha_j) &= \sigma_\alpha^2 \text{ if } i = j, \text{ 0 otherwise,} \\ E(\lambda_t \lambda_s) &= \sigma_\lambda^2 \text{ if } t = s, \text{ 0 otherwise,} \\ E(\eta_{it} \eta_{js}) &= \sigma_\eta^2 \text{ if } i = j, \text{ } t = s, \text{ 0 otherwise,} \\ E(\alpha_i \mathbf{X}_{it}) &= E(\lambda_t \mathbf{X}_{it}) = E(\eta_{it} \mathbf{X}_{it}) &= 0. \end{aligned}$$

If those assumptions are met, we can consider the “two-way variance components” model where:

$$\begin{aligned}\text{Var}(u_{it}) &= \text{Var}(Y_{it}|\mathbf{X}_{it}) \\ &= \sigma_{\alpha}^2 + \sigma_{\lambda}^2 + \sigma_{\eta}^2\end{aligned}$$

If we assume $\lambda_t = 0$, then we get a model like:

$$Y_{it} = \mathbf{X}_{it}\beta + \alpha_i + \eta_{it}$$

with total error variance:

$$\sigma_u^2 = \sigma_{\alpha}^2 + \sigma_{\eta}^2.$$

a.k.a. the “one-way error components model.”

“Random” Effects: Estimation

The model above will violate the standard OLS assumptions of uncorrelated errors, because the (compound) “errors” u_{it} within each unit share a common component α_i . Consider the within- i variance-covariance matrix of the errors $\mathbf{u}$:

$$\begin{aligned} E(\mathbf{u}_i \mathbf{u}_i') \equiv \boldsymbol{\Sigma}_i &= \sigma_\eta^2 \mathbf{I}_T + \sigma_\alpha^2 \mathbf{\bar{1}} \mathbf{\bar{1}}' \\ &= \begin{pmatrix} \sigma_\eta^2 + \sigma_\alpha^2 & \sigma_\alpha^2 & \cdots & \sigma_\alpha^2 \\ \sigma_\alpha^2 & \sigma_\eta^2 + \sigma_\alpha^2 & \cdots & \sigma_\alpha^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_\alpha^2 & \sigma_\alpha^2 & \cdots & \sigma_\eta^2 + \sigma_\alpha^2 \end{pmatrix} \end{aligned}$$

Assuming conditional independence across units, we then have:

$$\text{Var}(\mathbf{u}) \equiv \boldsymbol{\Omega} = \begin{pmatrix} \boldsymbol{\Sigma}_1 & 0 & \cdots & 0 \\ 0 & \boldsymbol{\Sigma}_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \boldsymbol{\Sigma}_N \end{pmatrix}$$

“Random” Effects: Estimation

Under our assumptions, we can show that:

$$\Sigma^{-1/2} = \frac{1}{\sigma_\eta} \left[\mathbf{I}_T - \left(\frac{\theta}{T} \mathbf{\ddot{u}}' \right) \right]$$

where

$$\theta = 1 - \sqrt{\frac{\sigma_\eta^2}{T\sigma_\alpha^2 + \sigma_\eta^2}}$$

is an unknown quantity to be estimated.

Estimation: Starting with an estimate of $\hat{\theta}$, calculate:

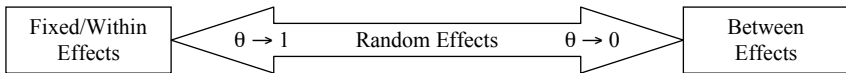
$$\begin{aligned} Y_{it}^* &= Y_{it} - \hat{\theta} \bar{Y}_i \\ X_{it}^* &= X_{it} - \hat{\theta} \bar{X}_i, \end{aligned}$$

then estimate:

$$Y_{it}^* = (1 - \hat{\theta})\alpha + X_{it}^* \beta_{RE} + [(1 - \hat{\theta})\alpha_i + (\eta_{it} - \hat{\theta} \bar{\eta}_i)]$$

and iterate between the two processes until convergence (a la FGLS).

“Random” Effects: An Alternative View



Random Effects

```
> RE<-plm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+log(GDPPerCapita)+NaturalResourceRents+
+         PostColdWar,data=WDI,effect="individual",model="random")
>
> summary(RE)
Oneway (individual) effect Random Effect Model
(Swamy-Arora's transformation)

Call:
plm(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +
     FertilityRate + log(GDPPerCapita) + NaturalResourceRents +
     PostColdWar, data = WDI, effect = "individual", model = "random")

Unbalanced Panel: n = 188, T = 8-51, N = 8252

Effects:
              var std.dev share
idiosyncratic 61.41   7.84  0.39
individual    95.61   9.78  0.61
theta:
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.727   0.877   0.888   0.881   0.888   0.888

Residuals:
      Min.  1st Qu.  Median    Mean 3rd Qu.    Max.
-33.9275  -5.3225  -0.5789  -0.0308   5.5470  46.1852

Coefficients:
              Estimate Std. Error z-value Pr(>|z|)
(Intercept)    -8.7087     2.4980  -3.49  0.00049 ***
PopGrowth       -0.1913     0.0824  -2.32  0.02024 *
UrbanPopulation  0.2267     0.0174  13.01 < 2e-16 ***
FertilityRate   -1.8838     0.1551 -12.15 < 2e-16 ***
log(GDPPerCapita) 7.0085     0.2744  25.54 < 2e-16 ***
NaturalResourceRents 0.0580     0.0167   3.48  0.00050 ***
PostColdWar     7.8811     0.2820  27.95 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares:    1070000
Residual Sum of Squares: 534000
R-Squared:              0.501
Adj. R-Squared:         0.501
Chsq: 8107.26 on 6 DF, p-value: <2e-16
```

A Nicer Table (Yet Again)

Table: Models of WBLI

	OLS	FE	BE	RE
Population Growth	-1.784*** (0.117)	-0.130 (0.081)	-4.989*** (1.114)	-0.191** (0.082)
Urban Population	-0.075*** (0.010)	0.337*** (0.018)	-0.067 (0.055)	0.227*** (0.017)
Fertility Rate	-2.990*** (0.155)	-1.548*** (0.155)	-1.546 (1.154)	-1.884*** (0.155)
ln(GDP Per Capita)	2.559*** (0.188)	8.569*** (0.289)	2.132* (1.112)	7.009*** (0.274)
Natural Resource Rents	-0.351*** (0.015)	0.090*** (0.017)	-0.300*** (0.091)	0.058*** (0.017)
Post-Cold War	9.882*** (0.366)	6.701*** (0.286)	6.547 (5.402)	7.881*** (0.282)
Constant	51.960*** (1.786)		57.340*** (12.600)	-8.709*** (2.498)
Observations	8,252	8,252	188	8,252
R ²	0.493	0.520	0.579	0.501
Adjusted R ²	0.493	0.508	0.565	0.501
F Statistic	1,337*** (df = 6; 8245)	1,454*** (df = 6; 8058)	41.5*** (df = 6; 181)	8,107***

*p<0.1; **p<0.05; ***p<0.01

“Random” Effects: Testing

Intuition:

- RE models require that $\text{Cov}(X_{it}, \alpha_i) = 0$.
- FE models do not.

This means that:

Model	<u>Reality</u>	
	$\text{Cov}(X_{it}, \alpha_i) = 0$	$\text{Cov}(X_{it}, \alpha_i) \neq 0$
Fixed Effects	Consistent, Inefficient	Consistent, Efficient
Random Effects	Consistent, Efficient	Inconsistent

The Hausman Test

Hausman test (FE vs. RE):

$$\widehat{W} = (\hat{\beta}_{\text{FE}} - \hat{\beta}_{\text{RE}})'(\hat{\mathbf{V}}_{\text{FE}} - \hat{\mathbf{V}}_{\text{RE}})^{-1}(\hat{\beta}_{\text{FE}} - \hat{\beta}_{\text{RE}})$$

$$W \sim \chi_k^2$$

Null: The RE model is consistent ($\text{Cov}(X_{it}, \alpha_i) = 0$).

Issues / Concerns:

- Asymptotic...
- No guarantee $(\hat{\mathbf{V}}_{\text{FE}} - \hat{\mathbf{V}}_{\text{RE}})^{-1}$ is positive definite
- This is a general specification test...

Hausman test (FE vs. RE):

```
> phptest(FE, RE) # ugh...
```

Hausman Test

```
data:  WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...  
chisq = 80, df = 6, p-value = 3e-15
```

```
alternative hypothesis: one model is inconsistent
```

Things to think about:

- Are we in asymptopia?
- Do we *believe* our model specification?...

Practical “Fixed” vs. “Random” Effects

Factors to consider:

- “Panel” vs. “TSCS” Data
- Nature of the Data-Generating Processes
- Importance of Non-Time-Varying Covariate Effects
- Dimension/Location of the Variance in the Data (N vs. T)

Connections: Hierarchical Linear Models

HLM Starting Points

Begin by considering a two-level “nested” data structure, with:

$$\begin{aligned} i &\in \{1, 2, \dots, N\} \text{ indexing first-level units, and} \\ j &\in \{1, 2, \dots, J\} \text{ indexing second-level groups.} \end{aligned}$$

A general two-level HLM is an equation of the form:

$$Y_{ij} = \beta_{0j} + \mathbf{X}_{ij}\beta_j + u_{ij} \tag{1}$$

where β_{0j} is a “constant” term, $\mathbf{X}_{ij}$ is a $NJ \times K$ matrix of K covariates, β_j is a $K \times 1$ vector of parameters, and $u_{ij} \sim \text{i.i.d. } N(0, \sigma_u^2)$ is the usual random-disturbance assumption.

Each of the $K + 1$ “level-one” parameters is then allowed to vary across Q “level-two” variables $\mathbf{Z}_j$, so that:

$$\beta_{0j} = \gamma_{00} + \mathbf{Z}_j\gamma_0 + \varepsilon_{0j} \quad (2)$$

for the “intercept” and

$$\beta_{kj} = \gamma_{k0} + \mathbf{Z}_j\gamma_k + \varepsilon_{kj} \quad (3)$$

for the “slopes” of $\mathbf{X}$. The ε s are typically assumed to be distributed multivariate Normal, with parameters for the variances and covariances selected by the analyst. Substitution of (2) and (3) into (1) yields:

$$Y_{ij} = \gamma_{00} + \mathbf{Z}_j\gamma_0 + \mathbf{X}_{ij}\gamma_{k0} + \mathbf{X}_{ij}\mathbf{Z}_j\gamma_k + \mathbf{X}_{ij}\varepsilon_{kj} + \varepsilon_{0j} + u_{ij} \quad (4)$$

The form is essentially a model with “saturated” interaction effects across the various levels, as well as “errors” which are multivariate Normal.

Model Assumptions

- Linearity / Additivity
- Normality of u s
- Homoscedasticity
- Residual Independence:
 - $\text{Cov}(\varepsilon_{\cdot j}, u_{ij}) = 0$
 - $\text{Cov}(u_{ij}, u_{i\ell}) = 0$

Model Fitting

- MLE
- “Restricted” MLE (“RMLE”)
- Choosing:
 - MLE is biased in small samples, especially for estimating variances
 - RMLE is not, but prevents use of LR tests when the models do not have identical fixed effects
 - In general: RMLE is better with small sample sizes, but MLE is fine in larger ones

Note that if we specify:

$$\beta_{0j} = \gamma_{00} + \varepsilon_{0j}$$

and

$$\beta_{kj} = \gamma_{k0}$$

we get:

$$Y_{ij} = \gamma_{00} + \mathbf{X}_{ij}\gamma_{k0} + \varepsilon_{0j} + u_{ij}$$

which is the RE model above (formally, a “one-level random-intercept” HLM).

In addition:

- HLMs can be expanded to 3- and 4- and higher-level models
- One can include cross-level interactions...

HLMs are widely used in education, psychology, ecology, etc. (less so in economics, political science). Also, there are many, many excellent [books](#), [websites](#), etc. that address HLMs

Random Effects Remix (using lmer)

```
> AltRE<-lmer(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+log(GDPPerCapita)+
  NaturalResourceRents+PostColdWar+(1|ISO3),data=WDI)
>
> summary(AltRE)
Linear mixed model fit by REML ['lmerMod']
Formula: WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate +
  log(GDPPerCapita) + NaturalResourceRents + PostColdWar + (1 | ISO3)
Data: WDI
```

REML criterion at convergence: 58449

Scaled residuals:

Min	1Q	Median	3Q	Max
-4.311	-0.653	-0.086	0.642	6.934

Random effects:

Groups	Name	Variance	Std.Dev.
ISO3	(Intercept)	348.9	18.68
Residual		61.5	7.84

Number of obs: 8252, groups: ISO3, 188

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	-21.5066	2.8297	-7.60
PopGrowth	-0.1481	0.0805	-1.84
UrbanPopulation	0.3000	0.0179	16.76
FertilityRate	-1.6537	0.1539	-10.75
log(GDPPerCapita)	8.0296	0.2825	28.42
NaturalResourceRents	0.0804	0.0165	4.87
PostColdWar	7.1147	0.2824	25.19

Correlation of Fixed Effects:

	(Intr)	PpGrwt	UrbnPop	FrtltR	l(GDPP	NtrlRR
PopGrowth	0.043					
UrbanPopltn	-0.180	0.034				
FertilityRt	-0.448	-0.225	0.416			
lg(GDPPPrCp)	-0.753	-0.060	-0.262	0.121		
NtrlRsrcRnt	0.001	-0.095	0.003	-0.108	-0.019	
PostColdWar	0.017	-0.029	-0.202	0.440	-0.125	-0.061

Q: Are They The Same? [A: More Or Less]

Table: RE and HLM Models of WBLI

	RE	AltRE
Population Growth	-0.191** (0.082)	-0.148* (0.081)
Urban Population	0.227*** (0.017)	0.300*** (0.018)
Fertility Rate	-1.884*** (0.155)	-1.654*** (0.154)
ln(GDP Per Capita)	7.009*** (0.274)	8.030*** (0.283)
Natural Resource Rents	0.058*** (0.017)	0.080*** (0.017)
Post-Cold War	7.881*** (0.282)	7.115*** (0.282)
Constant	-8.709*** (2.498)	-21.510*** (2.830)
Observations	8,252	8,252
R ²	0.501	
Adjusted R ²	0.501	
Log Likelihood		-29,224
Akaike Inf. Crit.		58,467
Bayesian Inf. Crit.		58,530
F Statistic	8,107***	

* p<0.1; ** p<0.05; *** p<0.01

For more, see [here](#).

Some reasons:

- Flexibility / verisimilitude
- Maximizing information (via pooling/shrinkage)
- Modeling / conceptual consistency

See especially Gelman and Hill (2007, Chapter 11).

Example: What if we think that the end of the Cold War had (slightly) different effects on WBLI in each country?...

HLM with Country-Level Random β s for ColdWar

```
> HLM1<-lmer(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+log(GDPPerCapita)+NaturalResourceRents+
PostColdWar+(PostColdWar|IS03),data=WDI,control=lmerControl(optimizer="bobyqa"))
```

```
> summary(HLM1)
```

```
Linear mixed model fit by REML ['lmerMod']
```

```
Formula: WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate +
log(GDPPerCapita) + NaturalResourceRents + PostColdWar +
(PostColdWar | IS03)
```

```
Data: WDI
```

```
Control: lmerControl(optimizer = "bobyqa")
```

```
REML criterion at convergence: 54996
```

```
Scaled residuals:
```

Min	1Q	Median	3Q	Max
-4.229	-0.540	-0.004	0.532	8.511

```
Random effects:
```

Groups	Name	Variance	Std.Dev.	Corr
IS03	(Intercept)	620	24.91	
	PostColdWar	140	11.83	-0.30
Residual		37	6.09	

```
Number of obs: 8252, groups: IS03, 188
```

```
Fixed effects:
```

	Estimate	Std. Error	t value
(Intercept)	-35.8452	3.3349	-10.75
PopGrowth	-0.2519	0.0647	-3.89
UrbanPopulation	0.3490	0.0204	17.13
FertilityRate	-3.6208	0.1678	-21.57
log(GDPPerCapita)	10.7794	0.3118	34.57
NaturalResourceRents	0.0235	0.0144	1.64
PostColdWar	2.3549	0.9958	2.36

```
Correlation of Fixed Effects:
```

	(Intr)	PpGrwt	UrbnPop	FrtltR	l(GDPP	NtrlRR
PopGrowth	0.038					
UrbanPopltn	-0.150	0.023				
FertilityRt	-0.509	-0.146	0.490			
lg(GDPPrCp)	-0.702	-0.062	-0.324	0.199		
NtrlRsrcRnt	0.019	-0.077	0.066	-0.048	-0.077	
PostColdWar	-0.216	0.002	-0.040	0.118	-0.008	-0.009

```
> anova(AltRE,HLM1)
refitting model(s) with ML (instead of REML)
Data: WDI
Models:
AltRE: WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate +
      log(GDPPerCapita) + NaturalResourceRents + PostColdWar + (1 | IS03)
HLM1: WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate +
      log(GDPPerCapita) + NaturalResourceRents + PostColdWar + (PostColdWar | IS03)
```

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
AltRE	9	58449	58512	-29215	58431			
HLM1	11	55004	55081	-27491	54982	3449	2	<2e-16 ***

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
> VarCorr(HLM1)
Groups   Name             Std.Dev. Corr
IS03      (Intercept) 24.91
          PostColdWar 11.83   -0.30
Residual                      6.09
```

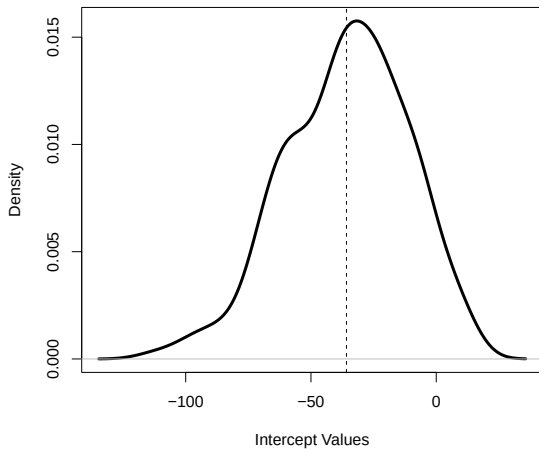
Random Coefficients

```
> Bs<-data.frame(coef(HLM1)[1])
>
> head(Bs)
  IS03..Intercept. IS03.PopGrowth IS03.UrbanPopulation IS03.FertilityRate
AFG           -24.48         -0.2519              0.349          -3.621
AGO           -34.10         -0.2519              0.349          -3.621
ALB           -22.00         -0.2519              0.349          -3.621
ARE          -111.68         -0.2519              0.349          -3.621
ARG           -78.19         -0.2519              0.349          -3.621
ARM           -30.82         -0.2519              0.349          -3.621
  IS03.log.GDPPerCapita. IS03.NaturalResourceRents IS03.PostColdWar
AFG                10.78                0.02354           3.430
AGO                10.78                0.02354          15.548
ALB                10.78                0.02354           5.880
ARE                10.78                0.02354           2.383
ARG                10.78                0.02354          23.435
ARM                10.78                0.02354           2.831

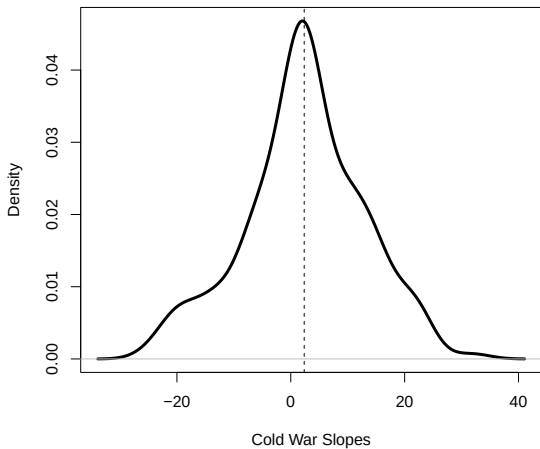
> mean(Bs$IS03..Intercept.)
[1] -35.85

> mean(Bs$IS03.PostColdWar)
[1] 2.355
```

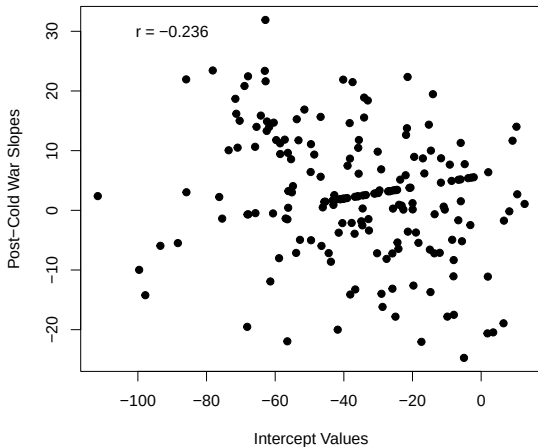
Random Intercepts (Plotted)



Random Slopes for Post-ColdWar (Plotted)



Scatterplot: Random Intercepts and Slopes



Hybrid Models: Combining Within and Between Effects

Recall that we can decompose the variance of our regression model as:

$$Y_{it} = \bar{\mathbf{X}}_i \beta_B + (\mathbf{X}_{it} - \bar{\mathbf{X}}_i) \beta_W + u_{it} \quad (5)$$

This raises the possibility of fitting the model in (5) directly...

- Simple to estimate (via OLS or other)
- Relatively easy interpretation
- Easy to test $\hat{\beta}_B = \hat{\beta}_W$

Example: Separate effects for within- and between-country *Natural Resource Rents*

- Theory: Countries with large natural resource endowments will have lower WBLI on average, *but*
- When natural resource rents increase *within* a country, that will be associated with higher expected WBLI...

Combining Within- and Between-Effects

Table: BE + WE Model of WBLI

	WEBE.OLS
Population Growth	-1.645*** (0.114)
Urban Population	-0.052*** (0.010)
Fertility Rate	-2.639*** (0.152)
ln(GDP Per Capita)	2.508*** (0.183)
Within-Country Nat. Resource Rents	0.119*** (0.027)
Between-Country Nat. Resource Rents	-0.528*** (0.017)
Post-Cold War	10.550*** (0.359)
Constant	50.480*** (1.745)
Observations	8,252
R ²	0.517
Adjusted R ²	0.517
Residual Std. Error	13.100 (df = 8244)
F Statistic	1,261.000*** (df = 7; 8244)

* p<0.1; ** p<0.05; *** p<0.01

Two important things to remember:

1. Recall the variation in *Natural Resource Rents*:

Variable	Dimension	Mean	SD	Min	Max	Observations
Natural Resource Rents	overall	7.284	11.184	0	88.592	N = 8252
	between		10.06	0	43.634	n = 188
	within		5.279	-23.94	61.391	T = 43.894

It is important to keep this in mind when discussing the relative sizes of effects:

- A one-s.d. increase in `NaturalResourceRents` within a country yields an expected change in Y of $5.3 \times 0.12 = 0.64$
- A one-s.d. difference in `NaturalResourceRents` between two countries yields an expected change in Y of $10.1 \times -0.53 = -5.4$

Interpretation (continued)

2. We can formally test whether $\hat{\beta}_B = \hat{\beta}_W$ for the *Natural Resource Rents* variable:

```
> library(car)
> linearHypothesis(WEBE.OLS,c("NRR.Within=NRR.Between"))
```

Linear hypothesis test:

NRR.Within - NRR.Between = 0

Model 1: restricted model

Model 2: WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate +
log(GDPPerCapita) + NRR.Within + NRR.Between + PostColdWar

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	8245	1485718				
2	8244	1415406	1	70312	410	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Extension: The “Mundlak Device”

Mundlak (1978)¹ notes that in:

$$Y_{it} = \mathbf{X}_{it}\beta + \alpha_i + u_{it}$$

the only portion of α_i that can be correlated with u_{it} must also be correlated with *only* the means of $\mathbf{X}$ for each individual. Thus, a model that includes / “controls for” the unit-level means of $\mathbf{X}_{it}$:

$$Y_{it} = \mathbf{X}_{it}\beta + \bar{\mathbf{X}}_i\gamma + u_{it}$$

...will ensure that the remaining variation in $\mathbf{X}_{it}$ is uncorrelated with u_{it} .

Moreover, a test for $\hat{\gamma} = 0$ is similar (and, as it turns out, superior) to the Hausman test for FE vs. RE.

¹Mundlak, Yair. 1978. “On the Pooling of Time Series and Cross Section Data.” *Econometrica* 46:69-85.

“Mundlak Device” (continued)

```
# Create unit-level means of *all* the (time-varying) predictors:
```

```
smol$PGBetween<-plm::Between(smol$PopGrowth,effect="individual")
smol$UPBetween<-plm::Between(smol$UrbanPopulation,effect="individual")
smol$FRBetween<-plm::Between(smol$FertilityRate,effect="individual")
smol$GDPBetween<-plm::Between(smol$lnGDPPerCap,effect="individual")
smol$NRRBetween<-plm::Between(smol$NaturalResourceRents,effect="individual")
smol$PCWBetween<-plm::Between(smol$PostColdWar,effect="individual")
```

```
# Regress Y on both using plain-old OLS:
```

```
MD<-plm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+lnGDPPerCap+
  NaturalResourceRents+PostColdWar+PGBetween+UPBetween+FRBetween+GDPBetween+
  NRRBetween+PCWBetween,data=smol,effect="individual",model="pooling")
```

FE and Mundlak Models

	FE2	MD
Population Growth	-0.130 (0.081)	-0.130 (0.127)
Urban Population	0.337*** (0.018)	0.337*** (0.029)
Fertility Rate	-1.548*** (0.155)	-1.548*** (0.245)
ln(GDP Per Capita)	8.569*** (0.289)	8.569*** (0.458)
Natural Resource Rents	0.090*** (0.017)	0.090*** (0.026)
Post-Cold War	6.701*** (0.286)	6.701*** (0.452)
Between-Country Population Growth		-5.327*** (0.246)
Between-Country Urban Population		-0.382*** (0.031)
Between-Country Fertility Rate		0.878*** (0.326)
Between-Country ln(GDP Per Capita)		-5.874*** (0.503)
Between-Country Nat. Resource Rents		-0.466*** (0.032)
Between-Country Post-Cold War		-0.511 (1.236)
Constant		49.820*** (2.381)
Observations	8,252	8,252
R ²	0.520	0.568
Adjusted R ²	0.508	0.568
F Statistic	1,454*** (df = 6; 8058)	903.8*** (df = 12; 8239)

* p<0.1; ** p<0.05; *** p<0.01

"Mundlak Device": Testing

```
> # Testing:
```

```
> linearHypothesis(MD,c("PGBetween = 0",  
+                        "UPBetween = 0",  
+                        "FRBetween = 0",  
+                        "GDPBetween = 0",  
+                        "NRRBetween = 0",  
+                        "PCWBetween = 0"))
```

Linear hypothesis test:

```
PGBetween = 0  
UPBetween = 0  
FRBetween = 0  
GDPBetween = 0  
NRRBetween = 0  
PCWBetween = 0
```

Model 1: restricted model

```
Model 2: WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate +  
  lnGDPPerCap + NaturalResourceRents + PostColdWar + PGBetween +  
  UPBetween + FRBetween + GDPBetween + NRRBetween + PCWBetween
```

```
Res.Df Df Chisq Pr(>Chisq)  
1    8245  
2    8239  6 1434    <2e-16 ***  
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Two-Way Unit Effects

Our original decomposition considered “two-way” effects:

$$Y_{it} = \mathbf{X}_{it}\boldsymbol{\beta} + \alpha_i + \eta_t + u_{it}$$

This “controls for” both non-time-varying unit-level differences and non-cross-sectionally-varying differences between periods. It also implies that we can use (e.g.) an F -test to examine the hypothesis:

$$H_0 : \alpha_i = \eta_t = 0 \quad \forall i, t$$

...that is, whether adding the (two-way) effects improves the model's fit. We can also consider the partial hypotheses:

$$H_0 : \alpha_i = 0 \quad \forall i$$

and

$$H_0 : \eta_t = 0 \quad \forall t$$

separately.

Two-Way Effects: Good & Bad

The Good:

- Addresses both time-invariant, unit-level heterogeneity and cross-sectionally-invariant, time-point-specific heterogeneity
- May be “fixed” or “random” ...
- Two-way FE is equivalent to differences-in-differences when $X \in \{0, 1\}$ and $T = 2$ (more on that later)

The (Potentially) Bad:

- Ability to control for unobserved heterogeneity depends on (probably invalid) functional form assumptions (Imai and Kim 2020)
- FE *requires* predictors that vary both within and between both units and time points
- RE requires the (additional) assumption that $\text{Cov}(\mathbf{X}_{it}, \eta_t) = \text{Cov}(\alpha_i, \eta_t) = 0$
- Two-way effects models ask a *lot* of your data (effectively fits $N + T + k$ parameters using NT observations)

Example: Two-Way Fixed Effects

```
> TwoWayFE<-plm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+log(GDPPerCapita)+NaturalResourceRents+
  PostColdWar,data=WDI,effect="twoway",model="within")
```

```
> summary(TwoWayFE)
Twoways effects Within Model
```

```
Call:
plm(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +
     FertilityRate + log(GDPPerCapita) + NaturalResourceRents +
     PostColdWar, data = WDI, effect = "twoway", model = "within")
```

```
Unbalanced Panel: n = 188, T = 8-51, N = 8252
```

```
Residuals:
```

```
      Min. 1st Qu.  Median 3rd Qu.    Max.
-32.056  -3.976   0.228   4.107  45.495
```

```
Coefficients:
```

	Estimate	Std. Error	t-value	Pr(> t)
PopGrowth	-0.2403	0.0672	-3.58	0.00035 ***
UrbanPopulation	0.0675	0.0160	4.23	2.4e-05 ***
FertilityRate	1.5960	0.1413	11.30	< 2e-16 ***
log(GDPPerCapita)	1.9740	0.2654	7.44	1.1e-13 ***
NaturalResourceRents	0.0528	0.0144	3.66	0.00025 ***

```
----
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Total Sum of Squares: 348000
```

```
Residual Sum of Squares: 339000
```

```
R-Squared: 0.0268
```

```
Adj. R-Squared: -0.00259
```

```
F-statistic: 44.1355 on 5 and 8009 DF, p-value: <2e-16
```

Two-Way Effects: Testing

```
> # Two-way effects:
```

```
> pFtest(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+
+       log(GDPPerCapita)+NaturalResourceRents+PostColdWar,data=WDI,
+       effect="twoway",model="within")
```

```
F test for twoways effects
```

```
data:  WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...
F = 115, df1 = 236, df2 = 8009, p-value <2e-16
alternative hypothesis: significant effects
```

```
> plmtest(TwoWayFE,c("twoways"),type=("kw"))
```

```
Lagrange Multiplier Test - two-ways effects (King and Wu)
```

```
data:  WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...
normal = 191, p-value <2e-16
alternative hypothesis: significant effects
```

```
> # One-way effects in the two-way model:
```

```
> plmtest(TwoWayFE,c("individual"),type=("kw"))
```

```
Lagrange Multiplier Test - (King and Wu)
```

```
data:  WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...
normal = 236, p-value <2e-16
alternative hypothesis: significant effects
```

```
> plmtest(TwoWayFE,c("time"),type=("kw"))
```

```
Lagrange Multiplier Test - time effects (King and Wu)
```

```
data:  WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...
normal = 92, p-value <2e-16
alternative hypothesis: significant effects
```

Two-Way Random Effects

```
> TwoWayRE<-plm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+log(GDPPerCapita)+NaturalResourceRents+
PostColdWar,data=WDI,effect="twoway",model="random")
```

```
> summary(TwoWayRE)
Twoways effects Random Effect Model
(Swamy-Arora's transformation)
```

Call:

```
plm(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +
FertilityRate + log(GDPPerCapita) + NaturalResourceRents +
PostColdWar, data = WDI, effect = "twoway", model = "random")
```

Unbalanced Panel: n = 188, T = 8-51, N = 8252

Effects:

	var	std.dev	share
idiosyncratic	42.338	6.507	0.31
individual	96.043	9.800	0.69
time	0.265	0.515	0.00

theta:

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
id	0.7715	0.8981	0.9074	0.9012	0.9074	0.9074
time	0.2335	0.2786	0.3136	0.2985	0.3186	0.3206
total	0.2329	0.2782	0.3131	0.2980	0.3181	0.3201

Residuals:

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	-66.04	-7.18	2.20	-0.24	10.61	35.67

Coefficients:

	Estimate	Std. Error	z-value	Pr(> z)
(Intercept)	-0.04796	0.32965	-0.15	0.88
PopGrowth	-0.20708	0.01028	-20.15	<2e-16 ***
UrbanPopulation	0.18181	0.00224	81.21	<2e-16 ***
FertilityRate	-0.68662	0.02010	-34.17	<2e-16 ***
log(GDPPerCapita)	5.47689	0.03580	153.00	<2e-16 ***
NaturalResourceRents	0.05609	0.00213	26.33	<2e-16 ***
PostColdWar	10.72261	0.04425	242.32	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 2930000

Residual Sum of Squares: 2320000

R-Squared: 0.267

Adj. R-Squared: 0.266

Chisq: 265840 on 6 DF, p-value: <2e-16

Table: Models of WBLI

	OLS	FE	BE	RE	TwoWayFE	TwoWayRE
Population Growth	-1.784*** (0.117)	-0.130 (0.081)	-4.989*** (1.114)	-0.191** (0.082)	-0.240*** (0.067)	-0.207*** (0.010)
Urban Population	-0.075*** (0.010)	0.337*** (0.018)	-0.067 (0.055)	0.227*** (0.017)	0.068*** (0.016)	0.182*** (0.002)
Fertility Rate	-2.990*** (0.155)	-1.548*** (0.155)	-1.546 (1.154)	-1.884*** (0.155)	1.596*** (0.141)	-0.687*** (0.020)
ln(GDP Per Capita)	2.559*** (0.188)	8.569*** (0.289)	2.132* (1.112)	7.009*** (0.274)	1.974*** (0.265)	5.477*** (0.036)
Natural Resource Rents	-0.351*** (0.015)	0.090*** (0.017)	-0.300*** (0.091)	0.058*** (0.017)	0.053*** (0.014)	0.056*** (0.002)
Post-Cold War	9.882*** (0.366)	6.701*** (0.286)	6.547 (5.402)	7.881*** (0.282)		10.720*** (0.044)
Constant	51.960*** (1.786)		57.340*** (12.600)	-8.709*** (2.498)		-0.048 (0.330)
Observations	8,252	8,252	188	8,252	8,252	8,252
R ²	0.493	0.520	0.579	0.501	0.027	0.267
Adjusted R ²	0.493	0.508	0.565	0.501	-0.003	0.266

*p<0.1; **p<0.05; ***p<0.01

Interpretation: modelsummary and marginaleffects

Two broadly useful packages by [Vincent Arel-Bundock](#):

1. `modelsummary`

- “modelsummary creates tables and plots to present descriptive statistics and to summarize statistical models in R.”
- In short, it... summarizes models (and data)

2. `marginalEffects`

- An “R package to compute and plot predictions, slopes, marginal means, and comparisons (contrasts, risk ratios, odds ratios, etc.) for over 70 classes of statistical models in R.”
- Focus is on individual covariates / predictors
- Especially handy for nonlinear models (GLMs, etc.)

Both are highly flexible and customizable, and **both play well with the models fit by `plm`.**

Using modelsummary

The most basic version of the code is:

```
> modelsummary(list(OLS,FE,RE))
```

...from which we get this (in the Viewer window):

	(1)	(2)	(3)
(Intercept)	51.963 (1.786)		-8.709 (2.498)
PopGrowth	-1.784 (0.117)	-0.130 (0.081)	-0.191 (0.082)
UrbanPopulation	-0.075 (0.010)	0.337 (0.018)	0.227 (0.017)
FertilityRate	-2.990 (0.155)	-1.548 (0.155)	-1.884 (0.155)
log(GDPPerCapita)	2.559 (0.188)	8.569 (0.289)	7.009 (0.274)
NaturalResourceRents	-0.351 (0.015)	0.090 (0.017)	0.058 (0.017)
PostColdWar	9.882 (0.366)	6.701 (0.286)	7.881 (0.282)
Num.Obs.	8252	8252	8252
R2	0.493	0.520	0.501
R2 Adj.	0.493	0.508	0.501
AIC	66288.4	57213.8	57837.4
BIC	66344.6	57262.9	57893.6
RMSE	13.42	7.74	8.04

Using modelsummary (continued)

A better-looking table:

```
> modelsummary(models,output="MS-Table-26.tex",title="Models of WDBI",stars=TRUE,fmt=2,  
+   gof_map=c("nobs","r.squared","adj.r.squared"),coef_rename=c("PopGrowth"="Population Growth",  
+   "UrbanPopulation"="Urban Population","FertilityRate"="Fertility Rate","log(GDPPerCapita)"="ln(GDP Per Capita)",  
+   "NaturalResourceRents"="Natural Resource Rents","PostColdWar"="Post-Cold War"))
```

Table 8: Models of WDBI

	OLS	Within	Random
(Intercept)	51.96*** (1.79)		-8.71*** (2.50)
Population Growth	-1.78*** (0.12)	-0.13 (0.08)	-0.19* (0.08)
Urban Population	-0.07*** (0.01)	0.34*** (0.02)	0.23*** (0.02)
Fertility Rate	-2.99*** (0.15)	-1.55*** (0.15)	-1.88*** (0.16)
ln(GDP Per Capita)	2.56*** (0.19)	8.57*** (0.29)	7.01*** (0.27)
Natural Resource Rents	-0.35*** (0.02)	0.09*** (0.02)	0.06*** (0.02)
Post-Cold War	9.88*** (0.37)	6.70*** (0.29)	7.88*** (0.28)
Num.Obs.	8252	8252	8252
R2	0.493	0.520	0.501
R2 Adj.	0.493	0.508	0.501

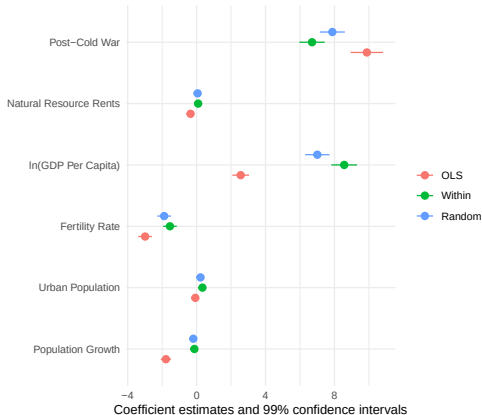
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

modelsummary: Coefficient Plots

Example code:

```
> modelplot(models, conf_level=0.99, coef_omit="(Intercept)", coef_rename=c("PopGrowth"="Population Growth",  
+ "UrbanPopulation"="Urban Population", "FertilityRate"="Fertility Rate",  
+ "log(GDPPerCapita)"="ln(GDP Per Capita)", "NaturalResourceRents"="Natural Resource Rents",  
+ "PostColdWar"="Post-Cold War"))
```

...from which we get:



Unit Effects Models: Software

R :

- the `plm` package; `plm` command
 - Fits one- and two-way FE, BE, RE models
 - Also fits first difference (FD) and instrumental variable (IV) models
- the `fixest` package; fast/scalable FE estimation for OLS and GLMs
- the `panelr` package (various commands)
- the `lme4` package; command is `lmer`
- the `nlme` package; command `lme`
- the `Paneldata` package

Stata : `xtreg`

- option `re` (the default) = random effects
- option `fe` = fixed (within) effects
- option `be` = between-effects
- Stata `package` `fect` = two-way models